Algorithm Design for Resilient Cyber-Physical Systems using an Automated Attack Generative Model

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***Abstract*—This paper presents a suite of algorithms for detecting and localizing attacks in cyber-physical systems, and performing improved resilient state estimation through a pruning algorithm. High performance rates for the underlying detection and localization algorithms are achieved by generating training data that cover large region of the attack space. An unsupervised generative model trained by physics-based discriminators is designed to generate successful false data injection attacks. Then the generated adversarial examples are used to train a multi-class deep neural network which detects and localizes the attacks on measurements. Next, a pruning algorithm is included to improve the precision of localization result and provide performance guarantees for the resulting resilient observer. The performance of the proposed method is validated using numerical simulations of a water distribution cyber-physical system.**

***Index Terms*—Resilient Control and Estimation, False Data Injection Attack, Generative Model, Multilayer Perceptron, Pruning algorithm**

I. Introduction

Modern cyber-physical critical infrastructures (CPCI) are facing fast-evolving cyber threats. Cyberattacks can strongly impact the operation of the CPCIs. For instance, a suspected cyber intruder took control of the Prykarpattyaoblenergo power system control center in western Ukraine in December 2015, leaving 230,000 people without electricity for up to 6 hours. That was the first time hackers have successfully targeted a country's power grid [1]. Also, cyber attacks in water systems have already become a reality. In 2015, 25 cyber attacks were disclosed in several water systems [2]. And recently in 2020, a malicious cyber-attack attempted to raise the chlorine level in Israel's water supply to a dangerous proportion [3]. Supervisory Control and Data Acquisition (SCADA) is a critical part of the CPCI that is highly susceptible to cyberattacks. It is responsible for collecting measurements from Remote Terminal Units (RTU) and sending them to the Control Centers (CC) to perform various functions such as contingency analysis, optimal planning, state estimation (SE), etc. The primary purpose of those functions is to maintain stable and secure operation of the CPCI.

SE is a core function in the CCs which monitors the system states using information collected from the SCADA systems and other measurement devices. SE can also detect and identify bad data that can cause significant error in the resulting estimate. The SE algorithm's accuracy depends on how pure and accurate the system measurements are. If any malicious or erroneous measurements pass through undetected, it could mislead the operators into making catastrophically wrong decisions. Numerous studies have shown that traditional SE algorithms are vulnerable to False Data Injection Attacks (FDIAs), where an intruder aims to hack multiple RTUs or even communication channels to insert fake measurements in order to misguide the CCs operational process [4]–[7]. Since FDIA was introduced in [4], different researchers have proposed various techniques for detection and localization, including diagnostic robust generalized potential [8], generalized likelihood ratio [9], fast Go-Decomposition (GoDec) [10], Markov chain [11], Bayesian detection with binary hypothesis [12], modified unbiased finite impulse response (UFIR) estimator [13], and cosine similarity matching scheme [14] and some residual-based approaches such as the Kullback Leibler distance method [15], unscented Kalman filter [16], χ^2 failure detector [17] and a residual-based localization scheme [18]. However, due to their dependence on the system model, the associated model uncertainty would affect their accuracy. Thus, if the FDIAs are designed appropriately, these model-based or residual-based detectors can easily be bypassed [4], [18]. Take for example, filter-based attack detector designs [13], [16]; the recovered attack signals could have deviations due to the uncertainties of the underlying model used in the filter design. Thus, if the attacker substitutes current meter measurements with historical readings following the same distribution, the Bayesian framework and generalized likelihood ratio approaches are unable to identify this FDIA [19].

To overcome the limitations of traditional residual-based bad data detection approaches, data-driven solutions based on machine learning algorithms have been widely adopted for

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detecting and localizing FDIAs due to their fast execution times and accurate results [20]–[26]. [20] utilized and tested various machine learning algorithms in detecting FDIAs. The results showed that machine learning algorithms can detect FDIAs accurately and faster. [21] proposed a deep neural network algorithm that can automatically detect and localize FDIAs in power systems. The proposed method can integrate the underlying graph topology of the grid and spatial correlations of its measurement data to jointly detect and localize the FDIAs in power systems. [26] proposed a hierarchy of neural networks with the capability of detecting novel attacks by training networks to understand the entire normal space instead of attack space.

In addition to attack detection and localization, resilient estimation approaches focus on maintaining the correctness of state estimates in the presence of adversarial attacks, which enables resilient operation of the CPSs [27]. Although attacks are possibly unbounded and can be designed to "fool" ℓ_2 observers [4], [17], the attack signals are sparse due to the resource limitation of the attackers. Based on these characteristics, the authors in [28] proposed an ℓ_1 minimization program for estimator design on linear systems and proposed an upper-bound on the total number of compromised measurements for guaranteed secure estimation. Some extension work considered robustness on resilient ℓ_1 estimation in presence of modeling error [29], worst-case estimation error bound in presence of bounded noise [30], and $\ell_1 - \ell_2$ estimation scheme [31]. And some iterative resilient observer designs has been proposed for determinant system, such as event-trigger Luenburger Observer [32] and Gramian-based estimator [33]. For stochastic system, several attack-resilient control strategies have also been studied [34], [35]. In addition, resilient estimation and control design has also been studied for nonlinear systems [36].

Furthermore, some enhanced resilient CPS designs have been studied to accommodate even worse adversarial condition. To push the limit on the number of compromised sensors below which secure estimation would be guaranteed, additional prior information could be used in the resilient estimation design. Existing results have used such information as measurement prior, support prior, and state prior: In [37], three forms of *state prior* are considered: *sparsity information*, $(\alpha, \overline{n}_0)$ *sparsity information*,, and *side information*. These are information generally unavailable to the malicious agent. In [35], the authors designed an improved resilient estimator, with input and state constraints, against actuator attacks. In our previous work [38]–[40], the resiliency of ℓ_1 based observers is enhanced by utilizing a measurement prior, generated by a pre-trained Gaussian process. In [41], a resilient unscented Kalman filter was proposed based on improved support prior. In [18], a weighted ℓ_1 observer is designed to exploit the support prior more rigorously.

Alternatively, improved resiliency of CPS could also be achieved by using more accurate attack detection and localization algorithms. For data-driven attack detectors, especially supervised learning-based detectors, exploiting more of the attack space and system nominal conditions could improve accuracy significantly [42]. This has motivated researchers to study attack data generation strategies for improving the detection/localization boundaries of the underlying data-driven algorithms. The goal of the attack generation problem is to generate feasible attacks that could elicit significant targeted disruption on system, while bypassing bad data detection. Model-based attack generation strategies often assume the attackers have the knowledge of system models [4], [17]. This assumption limits the application range of those attack generation strategies. Thus, automated attack generation without knowledge of the system models, is currently gaining a lot of attention. Although this topic has been studied widely in the machine learning area, for example; adversarial machine learning [43] and deepfake [44], the automated attack generation problem for CPS is a new challenging problem since the physical process dynamics is included. Inspired by the generative adversarial network (GAN) [45], some earlier researches [46]–[48] have studied GAN-based attack generation by using an existing attack dataset. The idea is to use GAN to learn the underlying distribution of feasible attacks, then a richer dataset of attacks could be generated. However, it is not practical to assume the existence of a non-generative attack dataset. Thus, an automated attack generation process, without attack examples, is needed. To the best of the authors knowledge, this paper is the first paper to study this problem.

Contribution: This paper proposes an enhanced resilient solution for CPSs subject to FDIAs. It is a hybrid, physics-based and data-driven, detection-based resilient estimation framework. The proposed framework consists of an automated attack generator, supervised learning-based attack detection and localization algorithms, pruning algorithms, and a state estimator. To train a supervised learning-based detector, abundant adversarial attack examples are often required. Model-based attack generators such as [18], [49] generally adopt optimization-based mechanisms which are computationally expensive, even NP-hard most times. Most traditional data-driven attack generators [46], [47], in literature, still require the attack examples with labels for training. In this paper, we propose an unsupervised generative model (GM) trained by multiple physics-based discriminators derived from the plant model. Secondly, due to the inherent uncertainty in the precision of data-driven detection and localization approaches, a pruning algorithm is proposed to systematically improve the precision of the resulting localization. The corresponding improvement guarantee is quantified analytically as a function of the aggressiveness of the pruning algorithm. Consequently, it is shown that the states of the system can be recovered correctly, thereby maintaining the operational performance of the CPS while attack is underway. Finally, the generalization of the proposed resilient design framework is demonstrated by showing how different state estimators and different supervised learning-based detectors could be placed in this frame.

The reminder of the paper is organized as follows: All the notations and the necessary mathematical tools used in the development are given in Section II. In Section III, the model of CPS is described. In Section IV, a formal definition of successful FDIAs is presented, and a data-driven generative model is proposed to generate successful FDIAs. In Section V, detection and localization approaches based on multi-class

MLP are given and a pruning algorithm is included to improve the resulting localization precision. In Section VI, a resilient ℓ_2 observer is designed to maintain correct state estimation under adversarial attacks. Conclusion remarks follow in Section VIII.

II. Notation and Preliminary

The following notations and definitions are used throughout the paper: $\mathbb{R}, \mathbb{R}^n, \mathbb{R}^{n\times m}$ denote the space of real numbers, real vectors of length n, and real matrices of n rows and m columns respectively. $\mathbb{R}_+$ denotes the space of positive real numbers. Normal-face lower-case letters $(e.g.\ x \in \mathbb{R})$ are used to represent real scalars, bold-face lower-case letters $(e.g.\ \mathbf{x} \in \mathbb{R}^n)$ represent vectors, while normal-face upper-case letters $(e.g.\ X \in \mathbb{R}^{n\times m})$ represent matrices. $X^\top$ denotes the transpose of the matrix X. The spectral radius of the square matrix X is denoted by $\rho(X)$. $X^\dagger \triangleq \left(X^\top X\right)^{-1} X^\top$ denotes the Moore-Penrose inverse. $\mathbf{1}_n$ and I_n denote the vector of ones and the identity matrix of size n respectively. Let $\mathcal{T} \subseteq \{1,\dots,n\}$, then, for a matrix $X \in \mathbb{R}^{m\times n}$, $X_\mathcal{T} \in \mathbb{R}^{|\mathcal{T}|\times m}$ is the sub-matrix obtained by extracting the rows of X corresponding to the indices in $\mathcal{T}$. $\mathcal{T}^c$ denotes the complement of a set $\mathcal{T}$, and the universal set on which it is defined will be clear from the context. We use I_T (or simply I when T is clear from context) to represent a T-time window $[i-T+1, i]$. In the same vein, $I-1$ is used to denote the time window $[i-T, i-1]$ accordingly. The support of a vector $\mathbf{x} \in \mathbb{R}^n$ is the set of the indices of nonzero entries in $\mathbf{x}$, defined as $\mathsf{supp}(\mathbf{x}) \triangleq \{i \subseteq \{1,\dots,n\} | \mathbf{x}_i \neq 0\}$. A vector $\mathbf{x} \in \mathbf{R}^n$ is said to be k-sparse if $|\mathsf{supp}(\mathbf{x})| \leq k$, and Σ_k denotes the subset of k-sparse vectors. A moving-horizon vector $\mathbf{x}_I \in \Sigma_k$ is given by $\mathbf{x}_I = \begin{bmatrix}\mathbf{x}_1^\top & \cdots & \mathbf{x}_T^\top\end{bmatrix} \in \mathbb{R}^{nT}$, where each constituent vector $\mathbf{x}_j \in \Sigma_k, j \in I$. $\mathsf{argsort}\downarrow(\mathbf{x})$ denotes a function that returns the sorted indices of vector $\mathbf{x}$ in descending order of the magnitude of $\mathbf{x}_i$. The symbol $\odot$ denotes element-wise multiplication of two vectors, as in $\mathbf{z} = \mathbf{x} \odot \mathbf{y}$, where $\mathbf{z}_i = \mathbf{x}_i\mathbf{y}_i$. The operator $\|\mathbf{z}\|_{1,\mathbf{w}} \triangleq \sum_{i=1}^n \mathbf{w}_i\mathbf{z}_i$ is a weighted 1-norm of a vector $\mathbf{z} \in \mathbb{R}^n$, with the weight vector $\mathbf{w} \in \mathbb{R}^n$. A continuous function $\phi : [0, a) \to [0, \infty)$ is said to belong to class $\mathcal{K}$ if it is strictly increasing and $\phi(0) = 0$. Both e^x and $\exp(x)$ are used to represent the exponential function. The symbol $*$ denotes the convolution operator of two vectors; given two vectors $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{u} * \mathbf{v} \triangleq \sum_j \mathbf{u}_j \mathbf{v}_{k+1-j}.$$

The symbol $\mathbb{E}$ denotes the expectation operator of a random variable. $z \sim \mathcal{B}(1,p)$ represents a Bernoulli distributed random variable z, with $\mathbb{E}[z] = p$. Thus, the sum of independent Bernoulli random variables $\{z_1, z_2, \cdots, z_N\}$ with $z_i \sim \mathcal{B}(1,p_i)$ satisfies the Poisson-binomial distribution, and the closed-form expression of the probability density function (pdf) of $\sum_i z_i$ is given by [50]:

$$\mathsf{Pr}\left\{\sum_{i=1}^N z_i = k\right\} = \mathbf{r}_k, \quad k = 0, \cdots, N, \tag{1}$$

where

$$\mathbf{r} = \prod_{i=1}^N p_i \cdot \begin{bmatrix}-s_1\\1\end{bmatrix} * \begin{bmatrix}-s_2\\1\end{bmatrix} * \cdots * \begin{bmatrix}-s_N\\1\end{bmatrix}, \tag{2}$$

with $s_i = -\frac{1-p_i}{p_i}$. The following lemma presents a result that will be useful for subsequent developments in this paper.

Lemma II.1. *For any non-decreasing convex function $\Phi : \mathbb{R} \to \mathbb{R}_+$ with $\Phi(0) = 1$,*

$$\mathbb{E}[\Phi(z)] \geq \mathsf{Pr}\{z > 0\}.$$

Proof. Based on the definition of expectation,

$$\begin{aligned}\mathbb{E}[\Phi(z)] &= \sum_{x\in\mathbb{R}} \Phi(x)\mathsf{Pr}\{z = x\}\\ &= \sum_{x\leq 0} \Phi(x)\mathsf{Pr}\{z = x\} + \sum_{x>0} \Phi(x)\mathsf{Pr}\{z = x\}.\end{aligned}$$

Since $\Phi(x) > 0$, then

$$\mathbb{E}[\Phi(z)] \geq \sum_{x>0} \Phi(x)\mathsf{Pr}\{z = x\}.$$

Also, since Φ is non-decreasing and $\Phi(0) = 1$, we have that $\Phi(x) \geqslant 1 \ \ \forall x > 0$. Thus,

$$\mathbb{E}[\Phi(z)] \geq \sum_{x>0} \mathsf{Pr}\{z = x\} = \mathsf{Pr}\{z > 0\}.$$

□

III. Model Development

The following model is used to describe the physical process of a CPS:

$$\begin{aligned}\mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i\\ \mathbf{y}_i &= C\mathbf{x}_i\end{aligned} \tag{3}$$

where $\mathbf{x}_i \in \mathbb{R}^n$ is the state vector, $\mathbf{u}_i \in \mathbb{R}^p$ the control input and $\mathbf{y}_i \in \mathbb{R}^{m_1}$ the sensor measurement. The appropriately dimensioned system matrices A, B and C are assumed to satisfy the following:

- A1. The pair (A, B) is controllable
- A2. There exists a positive integer $k_0 \leq m_1$ such that the pair $(A, C_\mathcal{S})$ is observable for all $\mathcal{S} \subset \{1, 2, \cdots, m_1\}$ with $|\mathcal{S}| \geq k_0$
- A3. $\rho\left(A^T\right) > 0$, where the integer T is a specified horizon parameter.

To estimate the system states from sensor measurements, the following receding horizon observer is considered:

$$\hat{\mathbf{x}}_i = A^T\mathbf{z} + F\mathbf{u}_{I-1}, \tag{4}$$

where

$$F = \begin{bmatrix}A^{T-1}B & A^{T-2}B & \dots & B\end{bmatrix},$$

and

$$\begin{aligned}\mathbf{z} &= \arg\min_{\mathbf{x}\in\mathbb{R}^n} \|\mathbf{y}_I - H_1\mathbf{u}_{I-1} - H_0\mathbf{x}\|_2\\ &= H_0^\dagger\mathbf{y}_I - H_0^\dagger H_1\mathbf{u}_{I-1},\end{aligned} \tag{5}$$

where

$$H_0 = \begin{bmatrix} CA \\ CA^2 \\ \vdots \\ CA^T \end{bmatrix}$$
$$H_1 = \begin{bmatrix} CB & 0 & \cdots & 0 \\ CAB & CB & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{T-1}B & CA^{T-2}B & \cdots & CB \end{bmatrix}. \quad (6)$$

Thus,

$$\hat{\mathbf{x}}_i = A^T H_0^\dagger \mathbf{y}_I + \left(F - A^T H_0^\dagger H_1\right) \mathbf{u}_{I-1}. \quad (7)$$

Consequently, the following control law is considered:

$$\mathbf{u}_i = K_p \hat{\mathbf{x}}_i, \quad (8)$$

where $\hat{\mathbf{x}}_i \in \mathbb{R}^n$ is the state estimate in (7), which is well-known (see [51] for example) to achieve $\lim_{i\to\infty} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2 = 0$, and the gain matrix K_p is designed to achieve $\rho\left(A + BK_p\right) \in (0,\ 1)$. Substituting (8) into the dynamics in (3) yields

$$\begin{aligned} \mathbf{x}_{i+1} &= A\mathbf{x}_i + BK_p\hat{\mathbf{x}}_i \\ &= \left(A + BK_p\right)\mathbf{x}_i + BK_p\left(\hat{\mathbf{x}}_i - \mathbf{x}_i\right), \end{aligned}$$

which implies that

$$\mathbf{x}_i = \left(A + BK_p\right)^i \mathbf{x}_0 + \sum_{j=0}^{i} \left(A + BK_p\right)^{i-j} BK_p \left(\hat{\mathbf{x}}_j - \mathbf{x}_j\right).$$

Thus,

$$\begin{aligned} \sum_{i=0}^{\infty} \|\mathbf{x}_i\|_2 \le& \left(\sum_{i=0}^{\infty} \rho\left(\left(A + BK_p\right)^i\right)\right) \|\mathbf{x}_0\|_2 \\ &+ \sum_{i=0}^{\infty}\sum_{j=0}^{i} \rho\left(\left(A + BK_p\right)^{i-j} BK_p\right) \|\hat{\mathbf{x}}_j - \mathbf{x}_j\|_2. \end{aligned}$$

Since $\rho(A + BK) \in (0, 1)$, then $\sum_{i=0}^{\infty} \rho\left((A+BK)^i\right) \le \sum_{i=0}^{\infty} \left(\rho(A+BK)\right)^i < \infty$[1]. Therefore, the resulting closed-loop dynamics satisfies

$$\sum_{i=0}^{\infty} \|\mathbf{x}_i\|_2 \le \alpha_0 \|\mathbf{x}_0\|_2 + \varphi\left(\sum_{i=0}^{\infty} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2\right)$$

for a positive constant $\alpha_0 < \infty$ and a class $\mathcal{K}$ functional φ. Since $\hat{\mathbf{x}}_i \to \mathbf{x}_i$, it follows that the origin of the closed-loop dynamics is asymptotically stable [52].

[1]For any $\beta \in (0,1)$, $\sum_{i=0}^{\infty} \beta^i = \lim_{N\to\infty} \sum_{i=0}^{N} \beta^i = \lim_{N\to\infty} \frac{1-\beta^{N+1}}{1-\beta} = \frac{1}{1-\beta} < \infty$

IV. Automated Attack Generation

In this section, a generative model is developed for generating successful FDIAs targeting the model in (3). Including attack signals in the CPS model in (3) yields the adversarial measurement model

$$\mathbf{y}_I = H_0 \mathbf{x}_{i-T} + H_1 \mathbf{u}_{I-1} + \mathbf{e}_I, \quad (9)$$

where $\mathbf{e}_I = \begin{bmatrix} \mathbf{e}_{i-T+1}^\top & \mathbf{e}_{i-T+2}^\top & \cdots & \mathbf{e}_i^\top \end{bmatrix}^\top$ contains the attack vectors within the moving window $[i - T + 1,\ i]$. Consequently, substituting (9) into (7) yields

$$\begin{aligned} \hat{\mathbf{x}}_i &= A^T \mathbf{x}_{i-T} + F\mathbf{u}_{I-1} + A^T H_0^\dagger \mathbf{e}_I \\ &= \mathbf{x}_i + A^T H_0^\dagger \mathbf{e}_I. \end{aligned} \quad (10)$$

Thus, the instantaneous estimation error is given by

$$\begin{aligned} \tilde{\mathbf{x}}_i &= \hat{\mathbf{x}}_i - \mathbf{x}_i \\ &= A^T H_0^\dagger \mathbf{e}_I. \end{aligned} \quad (11)$$

In general, attackers aim to achieve maximum perturbation of certain critical system output without triggering an alarm from the bad data detection mechanism. For this paper, a residual-based bad data detection scheme is used. Let

$$\mathbf{r}_I \triangleq \mathbf{y}_I - H_1 \mathbf{u}_{I-1} - H_0 \mathbf{z}, \quad (12)$$

where $\mathbf{z}$ is given in (5), be the residual vector associated with the receding horizon estimator in (7). Given a threshold value $\tau \in \mathbb{R}$, the bad data indicator for the windowed measurement history is given by

$$\mathsf{BDD}(\mathbf{y}_I) = \begin{cases} 0 & \text{if } \|\mathbf{r}_i\| \le \tau \\ 1 & \text{otherwise.} \end{cases} \quad (13)$$

Substituting (5) and (9) into (12) yields

$$\begin{aligned} \mathbf{r}_I &= \left(I - H_0 H_0^\dagger\right)\left(\mathbf{y}_I - H_1 \mathbf{u}_{I-1}\right) \\ &= \left(I - H_0 H_0^\dagger\right)\left(H_0 \mathbf{x}_{i-T} + \mathbf{e}_I\right). \end{aligned} \quad (14)$$

Thus, from the attacker's perspective, it is desirable to keep the quantity $\left\| \left(I - H_0 H_0^\dagger\right) \mathbf{e}_I \right\|_2$ as small as possible.

Next, let $\mathbf{y}_c = C_c \mathbf{x}$ be a vector of critical output targeted by the attacker. We remark that this is not an attempt to identify a priori the particular outputs targeted by an attacker. Rather, we postulate that there are critical functional outputs which the resilient system designer desires to protect against adversarial targeting. Thus, we define FDIAs which target those outputs specifically so that the resulting resilient observer can offer the desired protection. In addition, since most of existing attack generation approaches [4], [18] consider the deviation on state estimates as the attack effectiveness, the associated control law is often omitted in the attack generation problem. However, from the development above, it is necessary to consider the controller since the trajectory of the critical output vector $\mathbf{y}_c$ is also affected by the control law which is also affected by the compromised measurements. Substituting the control law in (8) into the linear model in (3) yields the closed loop system

$$\begin{aligned} \mathbf{x}_{i+1} &= \bar{A}\mathbf{x}_i + \bar{B}\tilde{\mathbf{x}}_i, \\ &= \bar{A}\mathbf{x}_i + \bar{B}A^T H_0^\dagger \mathbf{e}_I. \end{aligned} \quad (15)$$

where $\bar{A} = A + BK_p$ and $\bar{B} = BK_p$. Therefore,

$$\mathbf{y}_{c_{i+1}} = C_c\bar{A}\mathbf{x}_i + C_c\bar{B}A^T H_0^{\dagger}\mathbf{e}_I. \tag{16}$$

Thus, from the attacker perspective, it is desirable to keep the quantity $\left\|C_c\bar{B}A^T H_0^{\dagger}\mathbf{e}_I\right\|_2$ as big as possible.

We now have all the ingredients to formally define and synthesize the generative model for successful FDIAs in CPSs.

Definition 1 ((α,ϵ)-Successful FDIA)**.** *Consider the CPS in* (3)*. Given a positive integer $k < m$, the attack sequence $\mathbf{e}_I \in \Sigma_k \subset \mathbb{R}^{(T+1)m}$, is said to be (ϵ,α)-successful if*

$$\left\|C_c\bar{B}A^T H_0^{\dagger}\mathbf{e}_I\right\|_2 \geq \alpha \text{ and } \left\|\left(I - H_0H_0^{\dagger}\right)\mathbf{e}_I\right\|_2 \leq \epsilon. \tag{17}$$

for some $\alpha,\epsilon \in \mathbb{R}_+$.

Remark 1. *The first criterion above requires the effectiveness to be at least α, while the second criterion guarantees the stealthiness of the attack. Earlier researches formulated the attack generation problem as an effectiveness maximization problem with the stealthiness constraint [4], [17]. However, this limits the development of attack generators and its application range. Thus, feasibility-based definitions, as above, are currently favored by researchers [53].*

In the above definition, k quantifies the attack sparsity level per time. Specifically, it is the maximum number of attacks allowed at each time index. Examples of such successful FDIAs can be found in literature; a physics-based FDIA generator capable of fooling ℓ_2 observer, such as Kalman filer, can be found in [17] and an optimization-based FDIA generator capable of fooling stronger ℓ_1 observer was proposed in [18]. However, these FDIAs are either computationally too expensive or too conservative to implement on fast real-time systems. In order to circumvent these limitations, we drew inspiration from the well-known generative adversarial networks (GAN) [45] to build generative models for the elements of the set specified by inequalities in (17). The Definition 1 defines a target set of successful attack signals of the form:

$$\mathcal{S}(\alpha,\epsilon) \triangleq \left\{\mathbf{e} \mid \|M_1\mathbf{e}\|_2 \geq \alpha, \|M_2\mathbf{e}\|_2 \leq \epsilon\right\}. \tag{18}$$

Thus, the goal of the generative model is to learn how to generate elements of that set.

Definition 2. *Given a set $\mathcal{S} \subset \mathbb{R}^n$ and a prior distribution $P_z(\mathbf{z})$, a generative model is a mapping of the form $f : \mathbf{z} \mapsto \mathbb{R}^n$ such that $f(\mathbf{z}) \in \mathcal{S}$ with high probability.*

We consider a generative model of the form $G(\mathbf{z};\boldsymbol{\theta}_g)$, where $\boldsymbol{\theta}_g$ is a tunable parameter vector that will be prescribed to maximize the likelihood of generating elements in $\mathcal{S}$. Specifically, G is taken as a neural network whose input is a random number $\mathbf{z}$ sampled from a prior distribution P_z, with weights and biases contained in $\boldsymbol{\theta}_g$. Next, similarly to the GAN framework, $G(\mathbf{z};\boldsymbol{\theta}_g)$ is trained using a fixed discriminator defined by the boundary functions $\|M_1\mathbf{e}\|_2$ and $\|M_2\mathbf{e}\|_2$ of $\mathcal{S}$.

The training process is shown in Fig. 1. The output of the generator is passed, through the discriminator boundary

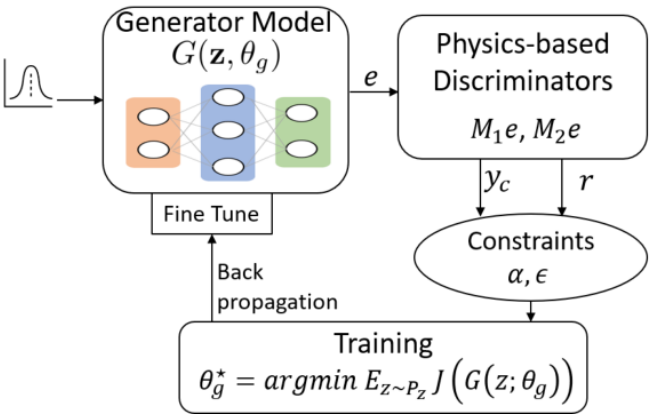


Fig. 1. Physics-based generative model and training process (α is the lower-bound constraint for attack effectiveness while ϵ is the upper-bound constraint for attack stealthiness)

functions, to a loss function designed such that the weights of the generator network are trained to ensure that the outputs of the discriminators satisfy the constraints. Since the constraints are threshold-based, we use an indicator-like loss function:

$$J(\mathbf{e}) = \mathsf{ReLU}(\alpha - \|M_1\mathbf{e}\|_2) + \mathsf{ReLU}(\|M_2\mathbf{e}\|_2 - \epsilon). \tag{19}$$

Consequently, the generator is trained by solving the unconstrained optimization problem

$$\boldsymbol{\theta}_g^* = \arg\min \mathop{\mathbb{E}}_{\mathbf{z}\sim\mathbf{P}_z} J(G(\mathbf{z};\boldsymbol{\theta}_g)) \tag{20}$$

via back propagation.

V. Detection and Localization

Through an automated attack generation by a well-trained generative model, an adversarial training dataset could be generated and used to improve the precision of the supervised learning-based attack detection and localization algorithm. In this section, a detection and localization algorithm is given by training a multi-layer perceptron (MLP) on the generated attack dataset. In addition, by analyzing the uncertainty of the MLP's output, a pruning algorithm is proposed to improve the localization precision further.

The term `attack location` refers to sensor channels (or indices) through which the attack signals are added to the system. Let $\mathcal{T}$ denote the actual attack support, then we define its indicator element-wise as:

$$\mathbf{q}_i \begin{cases} 0 & \text{if } i \in \mathcal{T} \\ 1 & \text{otherwise.} \end{cases} \tag{21}$$

Next, the definitions of detection and localization are given:

Definition 3 (Detection)**.** *Given a measurement vector $\mathbf{y} \in \mathbb{R}^m$, detection is a function $D : \mathbb{R}^m \to \{0,1\}$ whose output $D(\mathbf{y})$ indicates the existence of attack signal in the input $\mathbf{y}$.*

Definition 4 (Localization)**.** *Given a measurement vector $\mathbf{y} \in \mathbb{R}^m$, localization is a function $L : \mathbb{R}^m \to \{0,1\}^m$ whose output indicates detection result for each measurement channel. In particular, it is an estimate $\hat{\mathcal{T}}$ of the actual attack*

support $\mathcal{T}$. Its indicator $\hat{\mathbf{q}}$ is defined similarly to (21), using $\hat{\mathcal{T}}$

As universal function approximators, shown in Cybenko's theorem [54], MLPs are used to describe mathematical models via a regression scheme. Attack detection and localization is a multi-line binary classification task. With adequate training dataset generated by the automated attack generator, MLP's advantages of being a supervised learning network could be fully utilized.

The MLP used in this paper is a three-layer neural network containing two *LeakyReLU* hidden layers and one *Sigmoid* output layer, and the final classification outputs are 0 (for attack) if the corresponding MLP output is less than 0.5, and 1 (for no attack) otherwise. The output layer's size is the same as the input size and equal to the number of attacks. Consequently, the MLP is given by

$$g(\mathbf{y};\boldsymbol{\theta}) \triangleq \phi\left(\sum_{i=1}^{n} w_i \mathbf{y}_i + b\right), \tag{22}$$

where ϕ is the activation function of the output layer and $\boldsymbol{\theta}$ is a vector of tunable weights and biases. Then, the localization output is given by

$$\hat{\mathbf{q}}_i = \begin{cases} 1 & \text{if } g(\mathbf{y}_i;\boldsymbol{\theta}) \geq 0.5 \\ 0 & \text{otherwise.} \end{cases} \tag{23}$$

Next, a supervised learning algorithm, via back propagation, is used to train the MLP using the binary cross-entropy loss function:

$$J(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N}\Big(\mathbf{q}^{(i)}\log g(\mathbf{y}^{(i)};\boldsymbol{\theta}) + (\mathbf{1}-\mathbf{q}^{(i)};\boldsymbol{\theta})\log(\mathbf{1}-g(\mathbf{y}^{(i)};\boldsymbol{\theta}))\Big), \tag{24}$$

where $(\mathbf{y}^{(i)}, \mathbf{q}^{(i)})$ is the ith training data, and N is the total number of data points.

Since the MLP-based attack detection and localization method described above is a multi-line binary classification, the agreement between the localization result $\hat{\mathbf{q}}$ and the actual attack support $\mathbf{q}$ can be described by a Bernoulli distributed agreement variable $\epsilon_i \sim \mathcal{B}(1, \mathbf{p}_i)$ given by

$$\mathbf{q}_i = \epsilon_i \hat{\mathbf{q}}_i + (1-\epsilon_i)(1-\hat{\mathbf{q}}_i), \tag{25}$$

where $\mathbf{p}_i = \mathbb{E}[\epsilon_i] = \mathsf{Pr}\{\epsilon_i = 1\}$ is the confidence of the localization algorithm at i channel, which is often obtained by the true rate in *Receiver Operator Characteristic* (ROC) of the algorithm.

Consequently, the precision of the resulting support prior indicator $\hat{\mathbf{q}}$ is defined as follows:

Definition 5 (Positive Prediction Value, Precision, PPV [55])**.** *Given an estimate $\hat{\mathbf{q}} \in \{0,1\}^N$ of an unknown attack support indicator $\mathbf{q} \in \{0,1\}^N$, PPV is the proportion of $\mathbf{q}$ that is correctly identified in $\hat{\mathbf{q}}$. It is given by*

$$PPV = \frac{\|\mathbf{q} \odot \hat{\mathbf{q}}\|_0}{\|\hat{\mathbf{q}}\|_0}. \tag{26}$$

Next, with the attack localization result $\hat{\mathcal{T}}$, the estimated support of safe nodes can be represented as $\hat{\mathcal{T}}^c = \{j_1, j_2, \cdots, j_{|\hat{\mathcal{T}}^c|}\}$. A *Pruning Algorithm* is proposed to prune the most probable erroneous localization results in $\hat{\mathcal{T}}^c$.

The first step is to obtain a trust length $l_\eta (\leq Tm)$ based on the confidence of the localization results $\mathbf{p}$. Given a reliability level $\eta \in (0,1)$, the trust length l_η is an integer defined as the maximum number of safe nodes correctly localized in $\hat{\mathcal{T}}^c$ with a probability of at least η;

$$l_\eta \triangleq \max\left\{k \middle| \mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \geq k\right\} \geq \eta\right\}. \tag{27}$$

Let

$$\bar{\mathbf{r}} \triangleq \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{|\hat{\mathcal{T}}^c|} \end{bmatrix} = \prod_{i=1}^{|\hat{\mathcal{T}}^c|} \mathbf{p}(j_i) \begin{bmatrix} -s_{j_1} \\ 1 \end{bmatrix} * \begin{bmatrix} -s_{j_2} \\ 1 \end{bmatrix} * \cdots * \begin{bmatrix} -s_{j_{|\hat{\mathcal{T}}^c|}} \\ 1 \end{bmatrix}, \tag{28}$$

where $s_{j_i} = -\frac{1-\mathbf{p}_i}{\mathbf{p}_i}$. Then, $\mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = k\right\} = \bar{\mathbf{r}}_k$. Thus (27) becomes[2]

$$\begin{aligned} l_\eta &= \max\left\{k \middle| \mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \leq k\right\} \leq 1-\eta\right\} \\ &= \max\left\{k \middle| \sum_{i=0}^{k}\bar{\mathbf{r}}_i \leq 1-\eta\right\}. \end{aligned} \tag{29}$$

The second step is to obtain a *Pruned Support Prior* $\hat{\mathcal{T}}_\eta^c$ of size l_η. This is done through the robust extraction:

$$\hat{\mathcal{T}}_\eta^c = \left\{\mathsf{argsort}\downarrow(\mathbf{p}\odot g(\mathbf{y};\boldsymbol{\theta}))\right\}_1^{l_\eta}, \tag{30}$$

where $\{\cdot\}_1^{l_\eta}$ denotes an index extraction from the first elements to l_η elements. To obtain a pruned support prior consistent with the model assumptions in Section III, its size should satisfy $l_\eta \geq k_0$, where k_0 is a lower bound of $\hat{\mathcal{T}}_\eta^c$ such that $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is observable. This can be guaranteed by the proper choice of the reliability level η, as given in the following theorem.

Theorem V.1. *Given a measurement redundancy preservation parameter $k_0 > 0$, the size of pruned support prior satisfies $l_\eta \geq k_0$ if*

$$\eta \leq 1 - \sum_{i=0}^{|\hat{\mathcal{T}}^c|} e^{k_0-i}\bar{\mathbf{r}}_i, \tag{31}$$

where $\bar{\mathbf{r}}$ is given by (28), and $\hat{\mathcal{T}}^c$ is the estimated support of safe nodes from the attack localization algorithm.

[2] $\mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \leq k\right\} \leq 1-\eta$ is equivalent to $\mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = 0\right\} + \mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = 1\right\} + \cdots \mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = k\right\} \leq 1-\eta$ since k could only be integer.

Proof. According to (27), a sufficient condition for $l_\eta \geq k_0$ is

$$\mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \geq k_0\right\} \geq \eta,$$

which is equivalent to:

$$\mathsf{Pr}\left\{k_0 - \sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \leq 0\right\} \geq \eta$$

and

$$\mathsf{Pr}\left\{k_0 - \sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \geq 0\right\} \leq 1-\eta. \tag{32}$$

According to Lemma II.1, let $z = k_0 - \sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i$, and take $\Phi(z) = \exp(z)$ yields a sufficient condition for (32) as

$$\begin{aligned}&\mathbb{E}\left[\exp\left(k_0 - \sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i\right)\right] \leq 1-\eta,\\ \equiv\ & e^{k_0}\mathbb{E}\left[\exp\left(-\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i\right)\right] \leq 1-\eta. \end{aligned}\tag{33}$$

Since $\mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = k\right\} = \bar{\mathbf{r}}_k$, it follows that

$$\begin{aligned}\mathbb{E}\left[\exp\left(-\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i\right)\right] &= \sum_{k=0}^{|\hat{\mathcal{T}}^c|} e^{-k}\mathsf{Pr}\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = k\right\}\\ &= \sum_{k=0}^{|\hat{\mathcal{T}}^c|} e^{-k}\bar{\mathbf{r}}_k. \end{aligned}\tag{34}$$

Combining (33) and (34) yields

$$\sum_{i=0}^{|\hat{\mathcal{T}}^c|} e^{k_0-i}\bar{\mathbf{r}}_i \leq 1-\eta. \tag{35}$$

Finally, an upper bound on η is obtained as (31). □

Remark 2. *From* (31)*, it is clear that, necessarily, the upper bound should be strictly positive. This yields the following pruning requirement on the localization performance parameters:*

$$\sum_{i=0}^{|\hat{\mathcal{T}}^c|} e^{-i}\bar{\mathbf{r}}_i < e^{-k_0}. \tag{36}$$

If this holds, according to the assumption A2, with at least k_0 safe nodes remaining in the pruned support prior $\hat{\mathcal{T}}_\eta^c$, $(A, C_{\hat{\mathcal{T}}_\eta^c})$ will be observable. Thus the asymptotic convergence of the estimation error is guaranteed.

Moreover, it was shown in [18] that the precision PPV_η of the pruned support prior $\hat{\mathcal{T}}_\eta^c$ given by (30) achieves

$$\mathsf{Pr}\{\mathsf{PPV}_\eta = 1\} \geq \eta. \tag{37}$$

This indicates that, with the probability of at least η, the measurement nodes contained in $\hat{\mathcal{T}}_\eta^c$ are all safe, and $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is observable according to Remark 2. Thus, with the probability of at least η, exact state estimation can be achieved by using ℓ_2 observer designed based on $(A, C_{\hat{\mathcal{T}}_\eta^c})$. Finally, Algorithm 1 summarizes the attack detection and localization using automated attack generation and pruning operation.

Algorithm 1: Attack detection and localization with automated attack generation and pruning

I. **Offline: Dataset generation and training.**
1) Given time horizon T, prepare M_1, M_2 matrices for physics-based discriminators ← (14), (16).
2) Given the number of attacks $|\mathcal{T}|$, randomize attack support $\mathcal{T}$, and obtain its indicator $\mathbf{q}$ ← (21).
3) Given attack indexes α, ϵ, train the generative model (GM) by solving the optimization problem in (20).
4) Generate attack dataset $\{Y\}$ using the trained GM. Evaluate generated attack dataset via simulation.
5) Return the effective subset of attack dataset as the training dataset $\{Y, \mathbf{q}\}$.
6) Train the MLP in (22) using the generated dataset $\{Y, \mathbf{q}\}$ via backpropagation.

II. **Online: Localization and pruning**
1) Input real-time measurements $\mathbf{y}$, obtain the classification result $\hat{\mathbf{q}}$ ← (22), (23);
2) Obtain the pruned support prior $\hat{\mathcal{T}}_\eta^c(\hat{\mathbf{q}}_\eta)$ ← (29), (30).

Output: $\hat{\mathcal{T}}_\eta^c$ (pruned support prior)

VI. Resilient Estimator

In this section, the ℓ_2 observer in (7) is redesigned based on the pruned support generated by the Algorithm 1 in order to maintain secure state estimation subject to FDIA. The scheme of the resilient ℓ_2 observer is shown in Fig. 2. The attack detection and localization algorithm identifies the locations of attacks and return an estimate of the support of safe nodes $\hat{\mathcal{T}}^c$. Then the pruning algorithm is used to downselect to the safest subset of $\hat{\mathcal{T}}^c$. Finally, the downselected subset of measurements $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$ is used in the ℓ_2 observer.

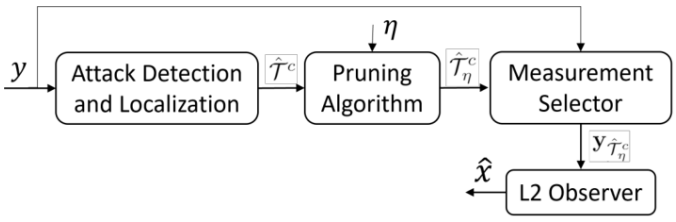


Fig. 2. Schematic diagram of resilient ℓ_2 observer

Given the pruned support $\hat{\mathcal{T}}_\eta^c$ within the moving window $[i-T+1, i]$, the resilient ℓ_2 observer is given by:

$$\hat{\mathbf{x}}_i^{'} = A^T H^{\dagger}_{0,\hat{\mathcal{T}}_\eta^c}\mathbf{y}_{\hat{\mathcal{T}}_\eta^c} + \left(F - A^T H^{\dagger}_{0,\hat{\mathcal{T}}_\eta^c} H_{1,\hat{\mathcal{T}}_\eta^c}\right)\mathbf{u}_{I-1}, \tag{38}$$

where $H_{0,\hat{\mathcal{T}}_\eta^c}, H_{1,\hat{\mathcal{T}}_\eta^c}$ are defined similarly to (6) using $C_{\hat{\mathcal{T}}_\eta^c}$ in place of C, and $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$ is part of the measurements indexed in

$\hat{\mathcal{T}}_\eta^c$. According to (37), $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$ contains only clean measurements with a probability of at least η. The next result gives the error bound of the resulting resilient estimator.

Theorem VI.1. *Given a pruned support $\hat{\mathcal{T}}_\eta^c$ from* (30)*. Suppose η satisfies* (31) *with the underlying attack localization algorithm satisfying the performance criterion in* (36)*. Then, with the probability of at least η, the resilient estimator in* (38) *achieves the same error performance as the attack-free estimates in* (7)*. Namely,*

$$\|\hat{\mathbf{x}}_i^{'} - \mathbf{x}_i^{\star}\|_2 = \|\hat{\mathbf{x}}_i - \mathbf{x}_i^{\star}\|_2 \tag{39}$$

where $\hat{\mathbf{x}}_i^{'}$ is the state estimate obtained using (38)*, $\hat{\mathbf{x}}_i$ is the attack-free estimate in* (7) *respectively, and $\mathbf{x}^\star$ is the actual state vector.*

Proof. We show that $\|\hat{\mathbf{x}}_i^{'} - \hat{\mathbf{x}}_i\|_2 = 0$ with probability of at least η. Since $\rho(A^T) > 0$, using the relationship $\hat{\mathbf{x}}_i = A^T\mathbf{z} + F\mathbf{u}_{I-1}$ and $\hat{\mathbf{x}}_i^{'} = A^T\mathbf{z}^{'} + F\mathbf{u}_{I-1}$, it is seen that $\|\hat{\mathbf{x}}_i^{'} - \hat{\mathbf{x}}_i\|_2 = 0$ is equivalent to $\|\mathbf{z}^{'} - \mathbf{z}\|_2 = 0$, where $\mathbf{z}$ and $\mathbf{z}^{'}$ are given by

$$\begin{aligned}\mathbf{z} &= H_0^\dagger \mathbf{y}_I - H_0^\dagger H_1 \mathbf{u}_{I-1} \\ &= H_0^\dagger H_0 \mathbf{x}_{i-T}\end{aligned} \tag{40}$$

and

$$\begin{aligned}\mathbf{z}^{'} &= H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger \mathbf{y}_{\hat{\mathcal{T}}_\eta^c} - H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger H_{1,\hat{\mathcal{T}}_\eta^c} \mathbf{u}_{I-1} \\ &= H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger H_{0,\hat{\mathcal{T}}_\eta^c} \mathbf{x}_{i-T} + H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger \mathbf{e}_{\hat{\mathcal{T}}_\eta^c}\end{aligned} \tag{41}$$

respectively. Since the pruned support prior is generated by (30), then (37) holds, which implies that, with the probability of at least η, $\mathbf{e}_{\hat{\mathcal{T}}_\eta^c} = \mathbf{0}$. Thus, (41) becomes

$$\mathbf{z}^{'} = H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger H_{0,\hat{\mathcal{T}}_\eta^c} \mathbf{x}_{i-T}. \tag{42}$$

Then combining (40) and (42) yields

$$\|\mathbf{z}^{'} - \mathbf{z}\|_2 = \left\| (H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger H_{0,\hat{\mathcal{T}}_\eta^c} - H_0^\dagger H_0)\mathbf{x}_{i-T} \right\|_2. \tag{43}$$

Since η satisfies (31) and the underlying attack localization algorithm satisfies the performance criterion in (36), then $|\hat{\mathcal{T}}_\eta^c| > k_0$ which implies $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is observable according to the assumption A2. Then, based on the definition of $H_{0,\hat{\mathcal{T}}_\eta^c}$ and H_0, they both have full rank. Therefore, since $k_0 < m_1$, the Moore-Penrose pseudo-inverses can be directly calculated as $H_{0,\hat{\mathcal{T}}_\eta^c}^\dagger = (H_{0,\hat{\mathcal{T}}_\eta^c}^\top H_{0,\hat{\mathcal{T}}_\eta^c})^{-1} H_{0,\hat{\mathcal{T}}_\eta^c}^\top$ and $H_0^\dagger = (H_0^\top H_0)^{-1} H_0^\top$. Then $\|\mathbf{z}^{'} - \mathbf{z}\|_2 = \|(I - I)\mathbf{x}_{i-T}\|_2 = 0$. □

By using ℓ_2 observer in (7), the attack-free state estimation error $\|\hat{\mathbf{x}}_i - \mathbf{x}_i^\star\|_2$ converges to zero as i goes infinity. According to (39), $\|\hat{\mathbf{x}}_i - \mathbf{x}_i^\star\|_2$ also goes to zero in the limit.

VII. Simulation

In this section, the corresponding improvement of resiliency due to the automated attack generation is shown in simulation, by implementing Algorithm 1 and the resilient observer in (38) on a water distribution system.

The water distribution system, shown in Fig. 3, contains 10 distributed operating tanks and 1 storage tank. The goal

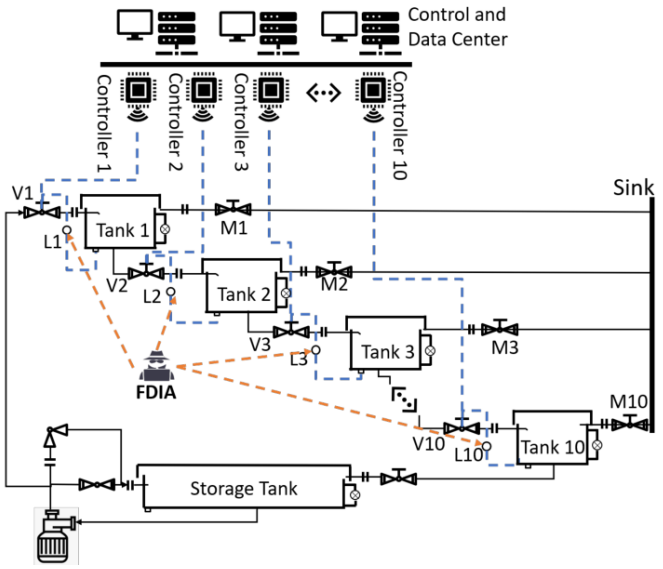


Fig. 3. Block diagram depiction of a water distribution system under FDIAs (black solid are water pipelines, blue dotted lines are wireless data transmission lines for sensors data and control commands, orange dotted lines are attack injection route)

is to regulate all operating tanks' water levels around desired values. The magnetic valves $V_1 - V_{10}$ at the inlet of operating tanks are controlled to adjust the water level of water tanks, and the manual valves $M_1 - M_{10}$ can be manipulated, at the demand point, to use the water from the corresponding tanks. The effect of these valves is modelled as the random demands $\mathbf{d}$ in (44). The valves at the storage tank are fixed at a constant opening value. For each operating tank, two sensors-water level meter and pressure sensors (L_1 L_{10}), are installed. The pressure sensors measure the difference of water level between adjoining tanks. The water levels adjustment process is approximated by the LTI model:

$$\begin{aligned}\mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i - \mathbf{d}_i, \\ \mathbf{y}_i &= C\mathbf{x}_i + \mathbf{v}_i + \mathbf{e}_i.\end{aligned} \tag{44}$$

where, $\mathbf{x}, \mathbf{u} \in \mathbb{R}^{10}$, $\mathbf{y}, \mathbf{v} \in \mathbb{R}^{61}$ are vector of water level, control input of magnetic valves, sensor measurements, and measurement noise respectively, $\mathbf{d} \in \mathbb{R}^{10}$ is the vector of random demands, and $\mathbf{e} \in \mathbb{R}^{61}$ is the injected FDIAs. For the sake of clean presentation, interested readers are directed to https://github.com/ZYblend/Resilient_WDS_AutoAttackGen for the values of system parameters $A \in \mathbb{R}^{10\times10}, B \in \mathbb{R}^{10\times10}, C \in \mathbb{R}^{61\times10}$. Given a desired water level $\mathbf{x}_d = 0.5 * \mathbf{1}_{10}$, the proportional controller below is used;

$$\mathbf{u}_i = -K(\hat{\mathbf{x}} - \mathbf{x}_d) - B^{-1}(A - I_{10})\mathbf{x}_d.$$

The critical measurement in this scenario is the sum of all water levels of operating tanks, thus, the critical measurement matrix is given by $C_c = [1\ 1\ \cdots\ 1] \in \mathbb{R}^{1\times10}$. Firstly, a simulation experiment is done to assess the observability of the simulated system in order to obtain the least number of measurements from which the system states are fully observable. A Monte Carlos experiment, with 50000 samples of randomly selected measurement subsets $\mathcal{S}$, is performed to calculated the

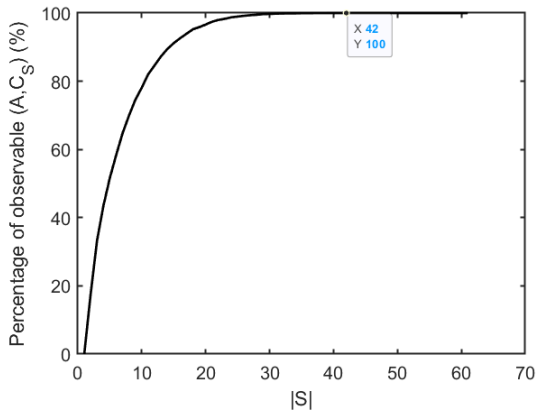


Fig. 4. Observability of simulated systems. The percentage of observable $(A, C_{\mathcal{S}})$ is calculated as the ratio of the number of observable $(A, C_{\mathcal{S}})$ to the total number of samples (=50000).

probability of full observerability of $(A, C_{\mathcal{S}})$ using $|\mathcal{S}|$. Fig 4 shows that full observerability is preserved with $k_0 \geq 42$, corresponding to at most 30% pruned measurements.

Next, we execute different simulation scenarios: automated attack generation, training data preparation with MLP training, and Monte-Carlo simulation for resilient ℓ_2 observer with different attack detection and localization strategies.

A. Automated attack generation

A fixed time step of $T_s = 0.01s$ is used and the moving-window size is set as $T = 8T_s$. With different numbers of attacks, the transfer matrices M_1, M_2 are calculated based on the corresponding random attack support. Then, the generator in (20) is trained as shown in Fig. 1. Due to the uncertainty in the discriminator caused by measurement noise and random demands, the threshold values are chosen from experience as $\alpha = 0.1, \epsilon = 0.5$. Fig. 5 shows the effectiveness and detection residual of the generated attacks during training process compared to the thresholds (α, ϵ). It is seen that the trained generator successfully transforms uniformly distributed inputs to feasible attacks satisfying the conditions in (18).

The trained generator is then used to generate a dataset $D_{\mathcal{S}}(\alpha, \epsilon)$ of attacks. In order to inject the generated attacks, recursive feasibility [56] should be considered. An online moving-horizon attack injection scheme used here is to solve the following sample-based optimization program at each time step i:

$$\begin{aligned} \underset{\mathbf{e}_i}{\text{Maximize}} \quad & \left\| M_1 \begin{bmatrix} \mathbf{e}^- \\ \mathbf{e}_i \end{bmatrix} \right\|_2 \\ \text{Subject to:} \quad & \left\| M_2 \begin{bmatrix} \mathbf{e}^- \\ \mathbf{e}_i \end{bmatrix} \right\|_2 \leq \epsilon \\ & \mathbf{e}_i \in D_{\mathcal{S}}(\alpha, \epsilon) \end{aligned} \tag{45}$$

where $\mathbf{e}^-$ is the attack history during the time window $I^- = [i - T + 1, i - 1]$. Fig. 6 shows the performance of the generated attacks in system operation through the injection program (45). The left figure shows that the T-horizon cumulative residual under attacks has small deviation from the nominal T-horizon cumulative residual, which means the attacks are hiding in the noise. However, the right figure

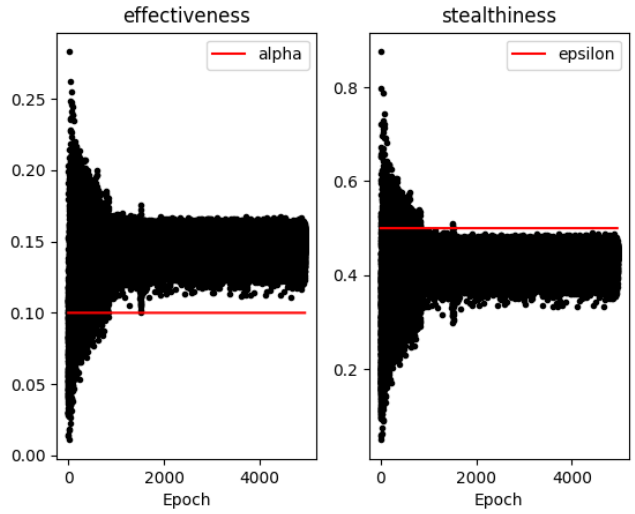


Fig. 5. Feasibility of generated attacks during training process. ϵ is the upper bound for the detection residual, α is the lower bound of effectiveness.

shows that the critical measurement has obvious deviation from nominal critical measurement after attacks are injected, which means the generated attacks inject targeted biases in the system successfully.

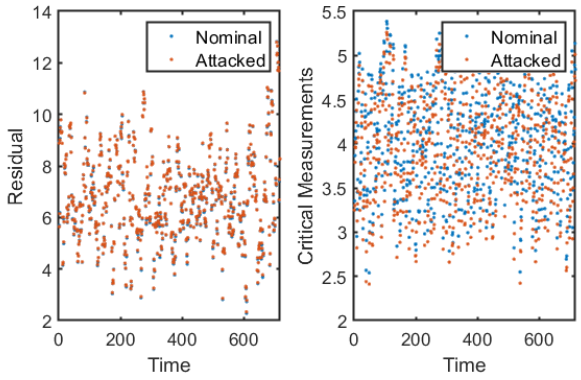


Fig. 6. Performance of generated attacks injected through 39% measurements nodes of the water distribution system. (Left: T-horizon cumulative residual, Right: critical measurement at each time step.)

Next, the above automated attack generation scheme is trained to generate different types of attacks including different attack supports and different number of attacks. The generated attacks are then used to train the supervised learning-based attack detection and localization algorithm in next subsection.

B. MLP training

To show one of the merits of the proposed automated attack generation scheme, two training datasets are prepared: one produced by random additive signals on measurements and the other augments the dataset with attack samples generated by the trained generator. Next, two MLP classifiers are trained on two datasets respectively. MLP1 and MLP2 are both four-layer fully-connected neural network with three "relu" hidden layers of $512, 1024, 512$ neurons respectively. The output layer has 61 neurons and uses "sigmoid" activation function. MLP1

is trained by the first random attack dataset, while MLP2 is trained the second training dataset containing automated generated attacks. The testing F1 score and precision of MLP1 are 85.1% and 99.7% respectively, while MLP2 achieves 88.7% and 99.1% respectively. The testing datasets used is 10% of the corresponding datasets, so the testing datasets for MLP2 and MLP1 are different. Next, we compare the localization precision of the two MLPs on a same testing dataset which is an equal mixture of dataset 1 and dataset 2. As shown in Fig. 7, MLP2 has successfully shifted the localization precision to the right, indicating a better performance overall. However, MLP1's performance is nearly uniform for different localization precision higher than 0.5. This is because it is trained by random attack dataset. The mean precision of MLP1 is 74.37%, and the mean precision of MLP2 is 98.69% on a testing set of successful FDIAs.

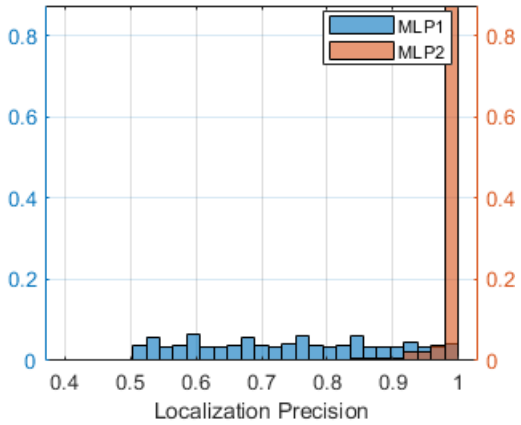


Fig. 7. A sample-based comparison of the performance of two MLP classifiers on same testing set of successful FDIAs. (MLP1: trained only with random attack generation, MLP2: trained with random attack generation and the proposed automated attack generation, 50000 test cases are used.)

As shown in Fig. 4, full observability can be guaranteed with certainty if $k_0 \geq 42$ measurements remains after pruning. For this, we evaluate the sufficient condition in (36) on MLP1 and MLP2. First, the two MLPs are run on 10000 cases generated by the trained generative model. From the results, the confidence vectors of the MLPs are estimated by calculating the true rates at all 61 measurement nodes; for node i, $\mathbf{p}_i = \frac{TP_i+TN_i}{P_i+N_i}$. Next, a sample based experiment is performed. For each support size $|\hat{\mathcal{T}}_j^c| \in \{40, 41, \ldots, 61\}$, we generated $\min\left\{{}^{61}C_{|\hat{\mathcal{T}}_j^c|}, 50000\right\}$ unique supports $\hat{\mathcal{T}}^c$, and evaluated the right hand side of (36) for each unique support as follows:

$$k_j = -\ln\left(\sum_{i=0}^{|\hat{\mathcal{T}}_j^c|} e^{-i}\bar{\mathbf{r}}_i\right),$$

where the Poisson-binomial probability density vector $\bar{\mathbf{r}}_i$ is calculated, using the confidence vectors of the corresponding MLP, based on (28). For each $|\hat{\mathcal{T}}_j^c| \in \{40, 41, \ldots, 61\}$, and each MLP, the sample probability of the event $k_j \geq k_0$ and the expected value of k_j are shown in Fig. 8. The results show that MLP2 has a bigger chance of preserving post-pruning observability. Indeed, according to the figure, post-pruning observability cannot be guaranteed when MLP1 is used. In addition, from the bottom plots in Fig. 8, it is seen that MLP2 reduces the uncertainty on the post-pruning observability compared to MLP1. By considering the sample-based analysis in Fig. 7 and Fig. 8 together, it is seen that MLP performs better when trained with data generated by the proposed automated attack generation.

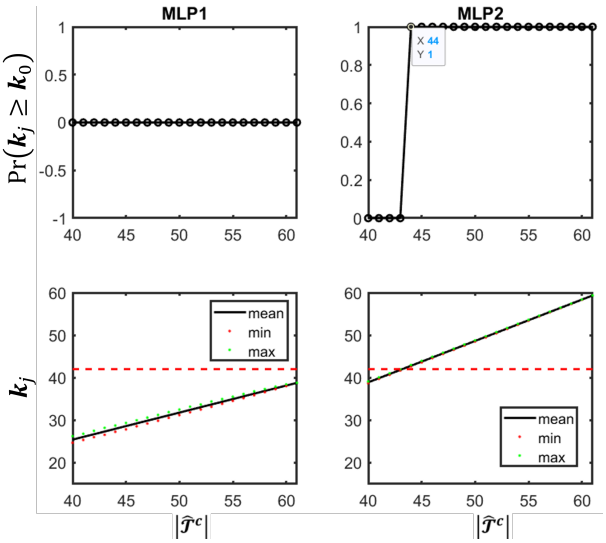


Fig. 8. A sample-based comparison of post-pruning observability for MLP1 and MLP2. (50000 test cases are used. The y axis for the top two subplots $Pr(k_j \geq k_0)$ is the sample estimate of the probability of at least k_0 measurements remaining after pruning. The black lines in the bottom subplots show the mean value of k_j that is the number of measurements remaining after pruning, while the red and green dots are the minimum and maximum values of k_j's respectively.

Next, we present a comparison of MLP1 and MLP2 in the resilient ℓ_2 estimation scheme for the water distribution system to show their effects on the estimator's resiliency.

C. Resilient ℓ_2 observer

Finally, the trained MLPs with pruning algorithm are used in a real-time simulation of the water distribution system. The resilient ℓ_2 observer in (38) is included to perform the state estimation. The goal of this simulation is to estimate the probability of achieving resilient performance, using different attack detection strategies with the ℓ_2 observer. A Monte-Carlo experiment is carried out. The input of the experiment are generated attacks by the trained attack generator. The outputs are: (1) the **probability of sustaining normal operation** (pSNO) estimated as the ratio of the instances where the deviation of critical measurement from the nominal value is below a given threshold to the total number of instances, (2) the **probability of triggering an alarm** (pTA) estimated as the ratio of the instances where the deviation of detection residual from the nominal value is beyond a given threshold to the total number of instances. Clearly, bigger values of pSNO and pTA indicate better resiliency.

The input attack dataset includes different number of attacks from 1 to 30, and for each number of attacks, we generated 5 datasets corresponding to different attack scenarios. The attack

supports were generated randomly. Next, the system simulation process shown in Fig. 2 was carried out with different attack detection strategies: MLP1, MLP1 with pruning, MLP2, and MLP2 with pruning. During the system simulation, the T-horizon cumulative detection residual $\mathbf{r}_I$ is calculated based on (14), and the critical measurement $\mathbf{y}_{c_{i+1}}$ is also calculated based on (16). Then the performance, in terms of deviation ratios of the residual $\Delta\mathbf{r}$ and the critical measurements $\Delta\mathbf{y}$ respectively, are evaluated using

$$\Delta\mathbf{r} = \frac{|(\mathbf{r}_I - \mathbf{r}_d) - (\mathbf{r}_I^\star - \mathbf{r}_d)|}{\max\left(|(\mathbf{r}_I - \mathbf{r}_d) - (\mathbf{r}_I^\star - \mathbf{r}_d)|\right)},$$

$$\Delta\mathbf{y} = \frac{\left|(\mathbf{y}_{c_{i+1}} - \mathbf{y}_{c_d}) - (\mathbf{y}^\star_{c_{i+1}} - \mathbf{y}_{c_d})\right|}{\max\left(\left|(\mathbf{y}_{c_{i+1}} - \mathbf{y}_{c_d}) - (\mathbf{y}^\star_{c_{i+1}} - \mathbf{y}_{c_d})\right|\right)},$$

where $\mathbf{r}_d$ is the residual vector in the corresponding attack-free and noise-free case, $\mathbf{y}_{c_d} = C_c\mathbf{x}_d$ is the desired critical measurement vector corresponding to the desired states $\mathbf{x}_d$. Consequently, pTA and pSNO are estimated as $\text{pTA} = \frac{1}{N}\left\|\Delta\mathbf{r} \geq \frac{\tau_1|\mathbf{y}_c^\star|}{\max(|\mathbf{y}_c - \mathbf{y}_c^\star|)}\right\|_{\ell_0}$ and $\text{pSNO} = \frac{1}{N}\left\|\Delta\mathbf{y} \leq \frac{\tau_2|\mathbf{r}^\star|}{\max(|\mathbf{r} - \mathbf{r}^\star|)}\right\|_{\ell_0}$ respectively, where N is the total number of time instances from the start of attack injection. The thresholds $\tau_1 = 1\%$ and $\tau_2 = 3\%$ are set based on the responses of the attack-free simulation.

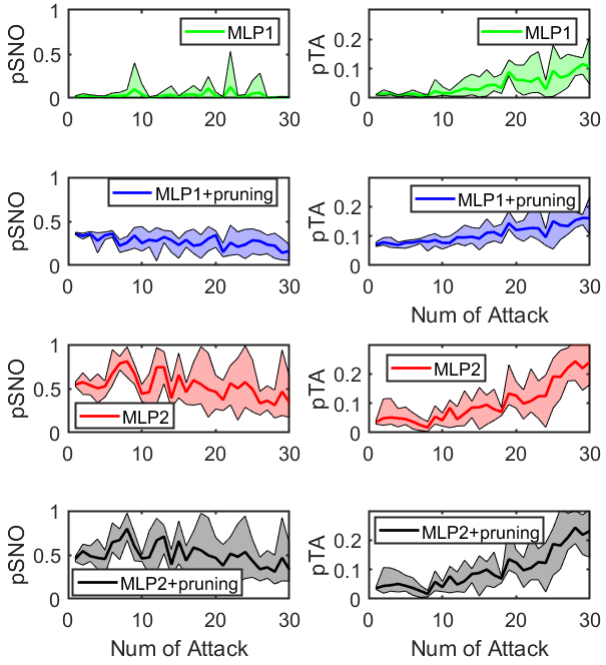


Fig. 9. A comparison of the ℓ_2 observer resiliency with different attack detection strategies, bigger values of pSNO and pTA indicate better resiliency. (The solid lines are mean values and the colored region indicates the spread of quantities

Fig. 9 shows the result of the experiment and a comparison of pSNO and pTA for different detection strategies. It is seen that MLP2 has better resiliency than MLP1, even better than MLP1 with pruning. This demonstrates the significant improvement in resiliency by including the automated attack generator in the training of MLP. In addition, the mean of localization precision, calculated by (26), of four attack detection strategies are shown in Fig. 10. As discussed in the last subsection, although the pruning algorithm still improve the localization precision of MLP2, the state estimation result is not improved as much due to the loss of observability. This indicates the MLP2 has already approached the best balance between the localization precision and the observability requirement. However, it is noteworthy that MLP2 with pruning has tighter spread of pSNO and pTA values than MLP2. This indicates that the pruning algorithm can reduce performance uncertainty.

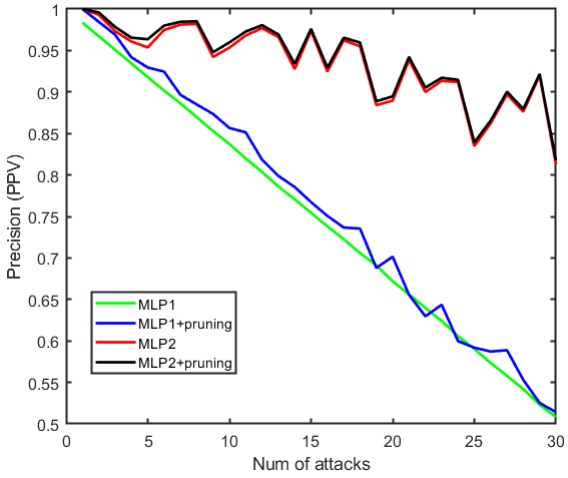


Fig. 10. A comparison of the mean precision of different attack detection strategies in the system simulation versus different number of attacks

Finally, a time-domain simulation result is presented in Fig. 11. The attacks, same as the ones in Fig 6, are injected through the program (45). Then four resilient estimation schemes are compared. It is seen that the schemes using MLP2 and MLP2 with pruning reduce the attack effectiveness on the critical measurements (smaller $\Delta\mathbf{y}$), and also reduce the attack stealthiness (bigger $\Delta\mathbf{r}$). This means that, with MLP2 or MLP2 and pruning, it is easier to detect the attacks while, at the same time, their effects are reduced.

VIII. Conclusion

In this paper, we present an algorithm design for resilient cyber-physical system, the resiliency is significantly improved by including the proposed automated attack generation in the training of attack detection models. Unlike traditional GAN-based FDIA generation, the automated attack generator does not require prior attack samples. This improved resilient estimation framework is a concurrent-learning based approach which contains a data-driven attack generator, an attack detection and localization algorithm, a model-based estimator design, and a pruning algorithm. The attack detection and localization algorithm used in this framework could be any supervised learning-based classification algorithm, not restricted to MLP.

There are several open questions to be addressed in future work: The discriminator is constructed based on the knowledge of the system dynamics, although the simulation in this paper

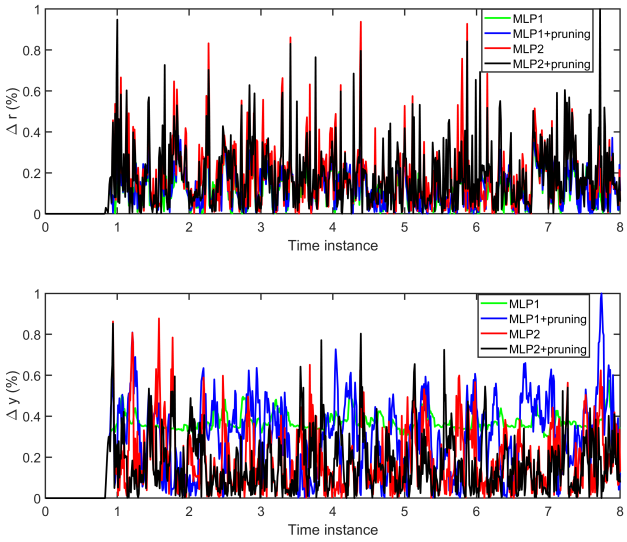


Fig. 11. A time-domain simulation results of the resilient estimation with four different attack localization strategies when the same 39% measurements are attacked as shown in Fig6

suggest that exact knowledge is not necessary for generating feasible attacks. The theoretical convergence proof of the generator with uncertainties on physics-based discriminator is still open. Data-driven discriminators with low computation burden is also a good problem to address. To improve the resiliency further, a weighted $\ell_1 - \ell_2$ state estimator could be used, but the theoretical development for bounded estimation error should be given with attack detector and pruning algorithm in the loop, as in Theorem V.1, and Theorem VI.1. Finally, the automated attack generation design could be further studied against the nonlinear systems.

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